

NIKHIL KRISHNAKUMAR PILLAI

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EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

June 2024

APJ Abdul Kalam Technological University, Kerala, India

SKILLS

Software Tools: Power BI, MATLAB, Arduino IDE, Microsoft Excel, Microsoft Word, Microsoft Outlook, Python (Pandas, NumPy, Matplotlib, PySpark, automation), SQL, Embedded C, C, C++.

Functional Skills: Embedded Software Development, splice location planning, retention & covering components, connector selection, DFMEA, DVP&R, DFM, DFA, prototype & pre-production build support, troubleshooting & rework, change management process, circuit design & analysis, microcontroller programming, sensor integration, PCB design & implementation, schematic reading, wireless communication, technical documentation, Cross-functional coordination, application-specific documentation, Build readiness support, Quality initiatives tracking.

WORK EXPERIENCE

Operations Management Trainee

August 2024 – Present

UPS, Howell, Michigan

- Coordinate with operations and logistics teams to manage inbound and outbound shipments with a focus on accuracy and quality standards.
- Support build readiness activities by unloading, stocking, and staging components to ensure timely availability for distribution.
- Maintain compliance with safety and inventory protocols while streamlining material flow across departments.

Project Coordinator

January 2025 – April 2025

Artic Traders

- Developed and implemented a procurement system for components using HTML, CSS, and PHP, enabling accurate tracking of BOMs and supplier specifications.
- Coordinated sourcing, purchasing, and component documentation in alignment with manufacturing requirements.
- Prepared KPI reports and cost tracking dashboards using Power BI and Excel, ensuring accurate program metrics and supporting change management activities.
- Supported cross-functional teams in change management and implementation of corrective actions for quality issues.

Data Scientist Intern

August 2024 – December 2024

GeeksforGeeks

- Built interactive Power BI dashboards to track engineering KPIs and warranty-related metrics, enabling data-driven quality initiatives.
- Performed data preprocessing, validation, and transformation using Python and SQL to ensure data accuracy, integrity, and support continuous process improvement initiatives.
- Documented and communicated continuous improvement plans, supporting task force meetings and program alignment.

Embedded Systems Project Trainee

August 2023 – September 2023

Revertech IT Solutions

- Developed real-time Arduino-based systems integrating multiple sensors and communication interfaces, demonstrating hardware integration and system troubleshooting skills.
- Assisted in layout planning for wiring and component placement to ensure reliability and ease of assembly.
- Prepared technical documentation for hardware integration and design improvements, supporting internal design reviews.

PROJECTS

Automatic Accident Detection System

- Designed and implemented an in-vehicle safety device using multi-sensor impact threshold analysis to detect collisions.
- Integrated GSM module for GPS-enabled emergency alerts and developed Embedded C firmware on Arduino, validated through simulated crash testing over a 6-month cycle.
- Applied design-for-assembly principles to ensure harness and sensor placement allowed ease of installation and troubleshooting.

NRF-based TEXTCOM System

- Developed a wireless campus alert system using NRF modules, custom PCBs, and microcontrollers, incorporating low-latency, bidirectional communication lines.
- Performed PCB design, soldering, assembly, and electrical testing; defined component retention methods to ensure mechanical stability and signal reliability.
- Integrated microcontrollers, NRF modules, and display interfaces into a complete system to optimize build efficiency.